

Lab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

Name: WANG Haoyu

Student ID: 22102084d

Part 2: Configure and Verify OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table?

show ip route ospf

Part 4: Configure OSPF Passive Interfaces

- g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? Serial0/0/0

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? 129

Does R2 show up as an OSPF neighbor on R1? Yes

Does R2 show up as an OSPF neighbor on R3? No

What does this information tell you?

R2 only adjacencies with R1, not R3 (due to passive S0/0/1 on R2). R3 routes to 192.168.2.0/24 via R1 indirectly, with higher cost 129 (64+64+1).

- h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

router ospf 1

no passive-interface s0/0/1

- i. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? Serial0/0/1

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

65. Cost of R3's s0/0/1 interface + the R2's g0/0 interface = 64+1 = 65.

Is R2 listed as an OSPF neighbor to R3? Yes

Part 5: Change OSPF Metrics

Step 1: Change the reference bandwidth on the routers.

- a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
```

```
R1(config-router)# auto-cost reference-bandwidth 100
```

```
% OSPF: Reference bandwidth is changed.
```

Please ensure reference bandwidth is consistent across all routers.

Why would you want to change the OSPF default reference-bandwidth?

The default 100Mb/s setting treats all faster links the same way, giving them a cost of 1, so you can't tell apart things like Fast Ethernet from Gigabit or 10G speeds.

Step 2: Change the bandwidth for an interface.

g.

Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

For 192.168.3.0/24, it's 781 from R1's S0/0/1 plus 1 from R3's G0/0, totaling 782. For 192.168.23.0/30, it's 781 from R1's S0/0/1 plus 781 from R3's S0/0/1, adding up to 1562.

- i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

~~1562~~. That's because every serial link is tuned to 128Kb/s bandwidth, bumping each one's cost to 781 based on the 100Mb/s reference divided by that speed, and the route goes through two such links.

Step 3: Change the route cost.

c.

Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

The direct path through S0/0/1 ends up costing 1566 overall (1565 plus 1), which is higher than the detour through R2 at 1563 (781 plus 781 plus 1). OSPF always picks the cheaper option.

Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?

Controlling it keeps things stable and unique in the OSPF setup, which helps a ton with managing logs, spotting issues, and avoiding ID shifts if interfaces change.

2. Why is the DR/BDR election process not a concern in this lab?

This lab sticks to point-to-point serial connections, which aren't the multiaccess type that needs DR or BDR elections to handle traffic efficiently.

3. Why would you want to set an OSPF interface to passive?

It stops pointless OSPF chatter on LAN sides where no other routers are connected, saving bandwidth, easing CPU strain, and boosting security by not spilling updates to random hosts.

4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?

It's the area structure that divides the network into zones, cutting down on update floods and allowing route summaries to keep things efficient as the network grows huge.

5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?

network 10.0.0.0 0.255.255.255 area 0

6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?

It'll go with EIGRP since its administrative distance is the lowest at 90, beating OSPF's 110 and RIP's 120 – the router checks that before even looking at metrics.